The University of Queensland Australia | CREATE CHANGE

Introduction to Token Reduction and Information Bottleneck

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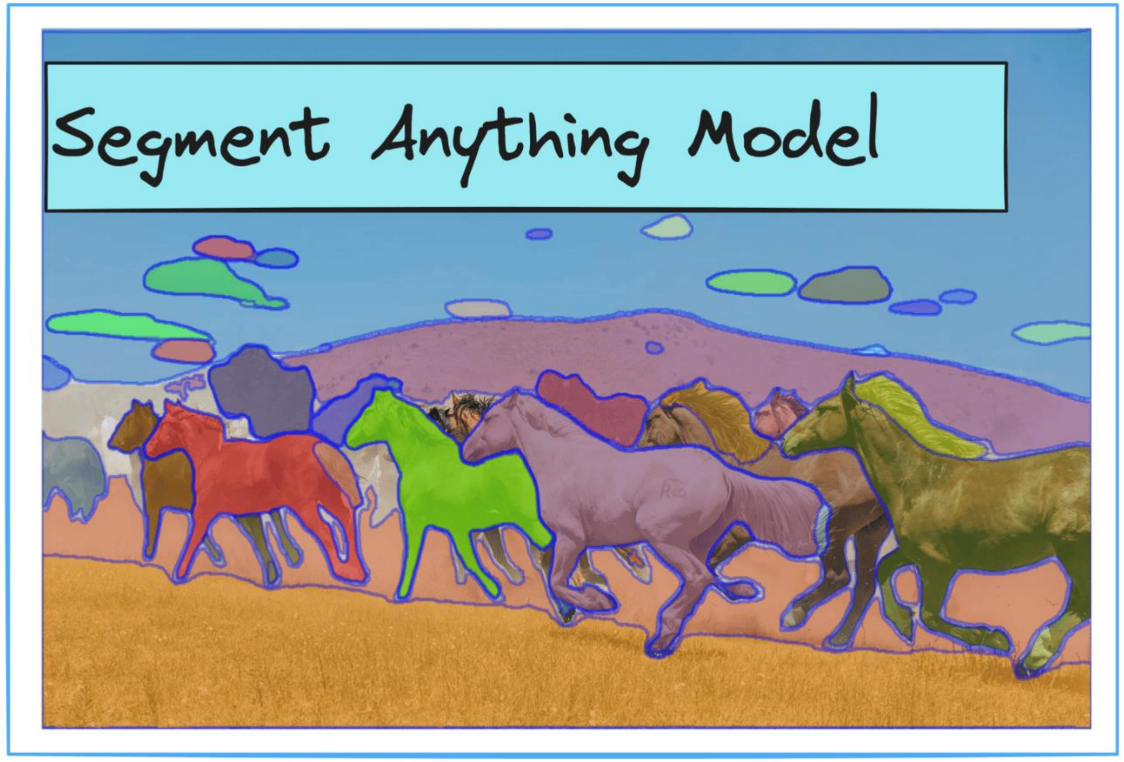
14/04/2026

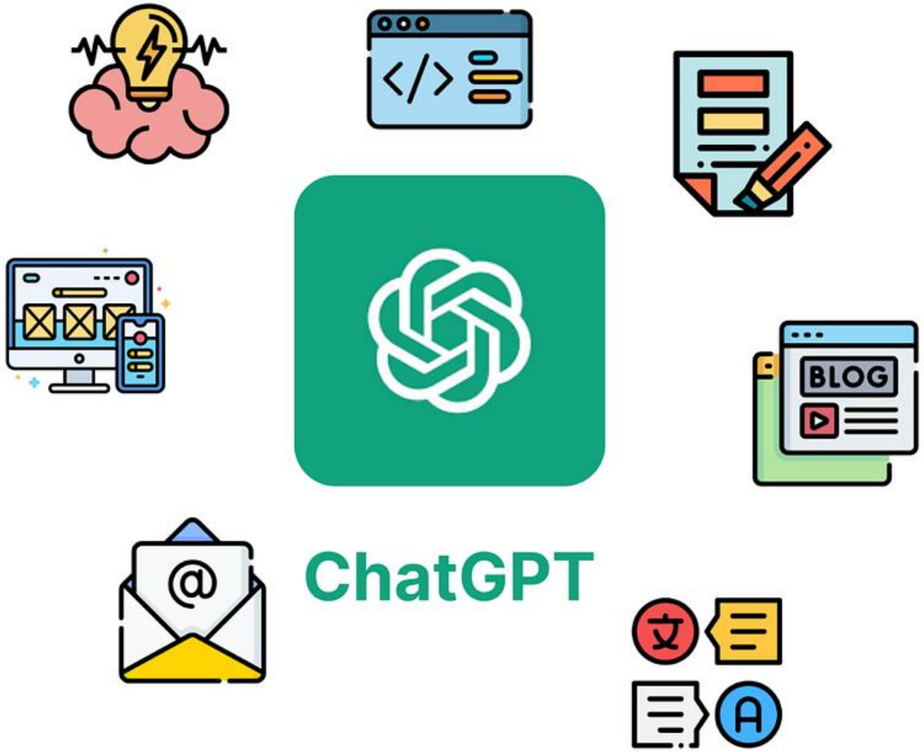
Introduction

Background

- Large pretrained models have significantly changed the AI research and our daily life.



Segment Anything (SAM)[1]



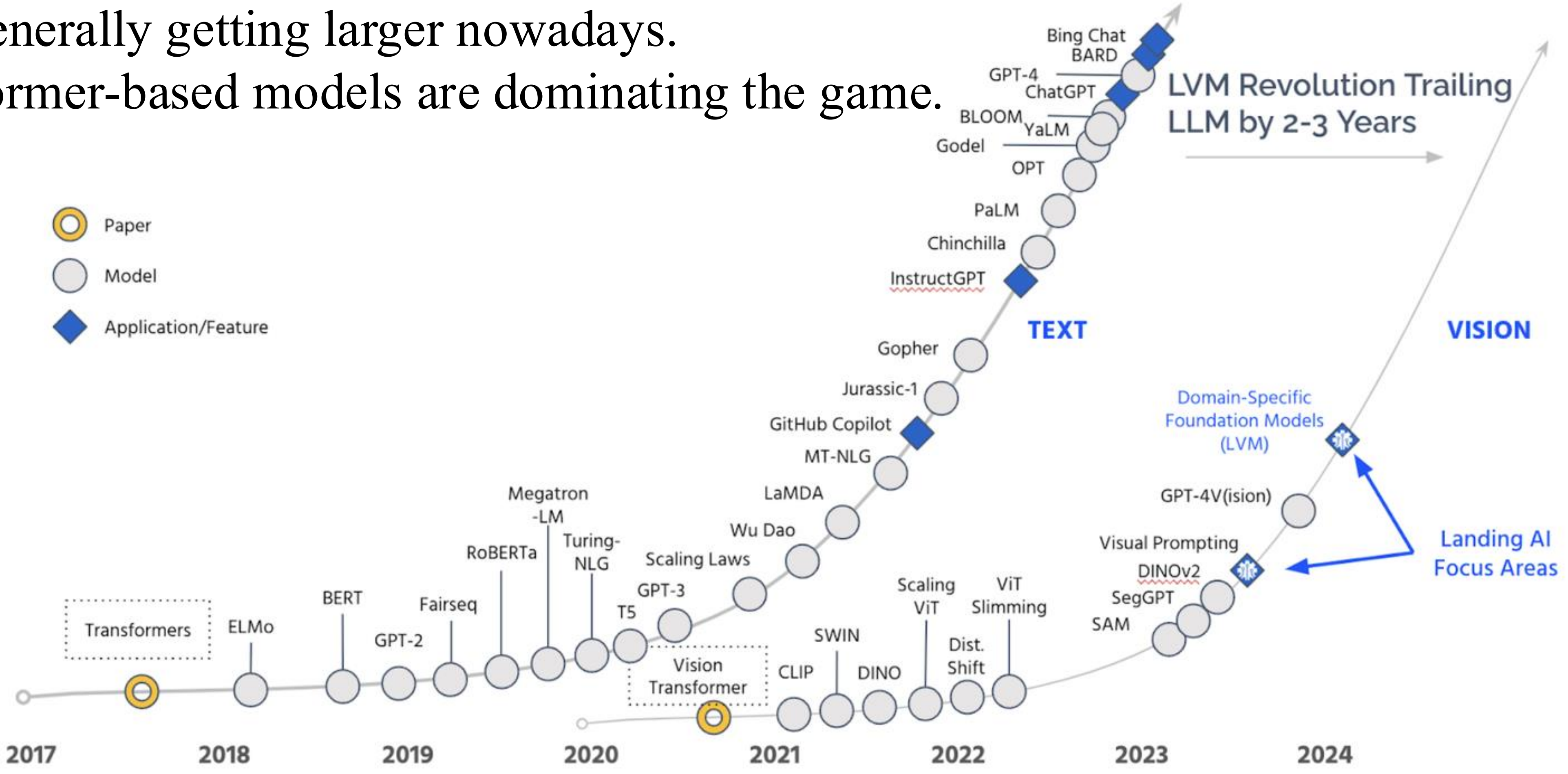
ChatGPT[2]

[1] Segment Anything picture from https://segment-anything.com/. Kirillov, Alexander, et al. "Segment anything." In ICCV. 2023.
[2] ChatGPT picture from https://www.flaticon.com/.

Introduction

Background

- Models are generally getting larger nowadays.
- Large Transformer-based models are dominating the game.



Model evolution during the past years[3]

[3] Picture from https://www.eetimes.com/andrew-ng-the-ai-text-revolution-is-coming-to-images/.

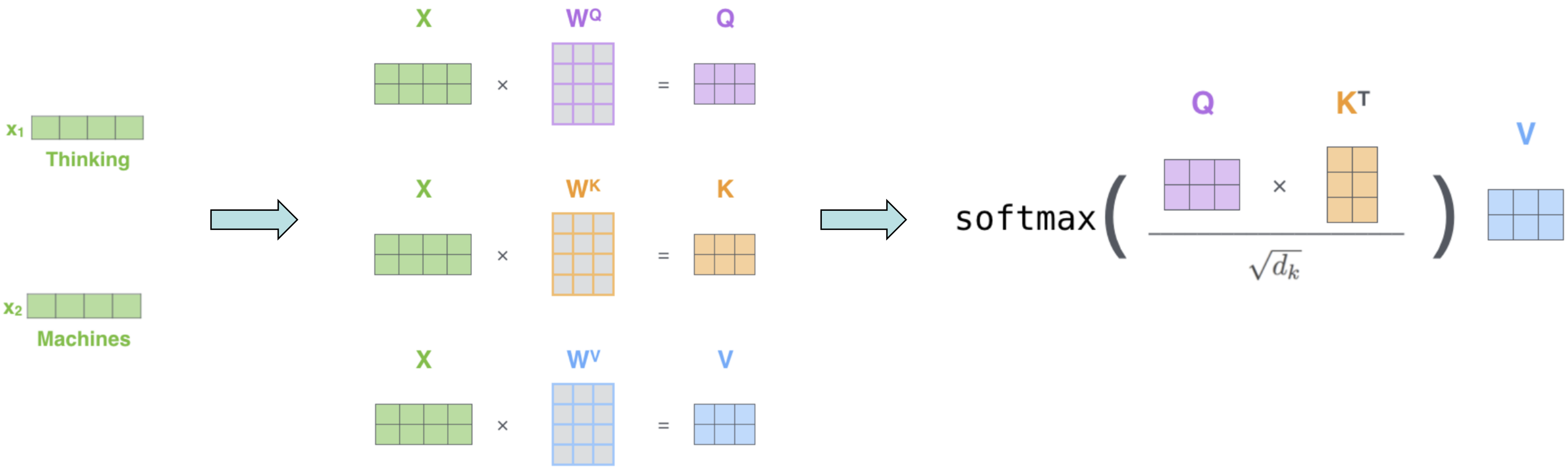
Introduction

Transformer and Self-Attention Mechanism[4]

- Calculate the attention weights by the input elements themselves

$$K = X_i W_K,\ Q = X_i W_Q,\ V = X_i W_V$$

$$X_{i+1} = \text{Attention}(K, Q, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$



[4] Vaswani, Ashish, et al. "Attention is all you need." In NeurIPS, 2017.

Introduction

Vision Transformer[5]

1. Split image into N image tokens
2. Perform Self-Attention on the image tokens

$$K = X_l W_K, \quad Q = X_l W_Q, \quad V = X_l W_V$$

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Self-attention mechanism in vision

[5] Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." In ICLR, 2021.

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Challenge:

- Complexity: $\mathrm{O}(2N^2C + 4NC^2)$
- Memory cost: $4NC + 4C^2 + hN^2$

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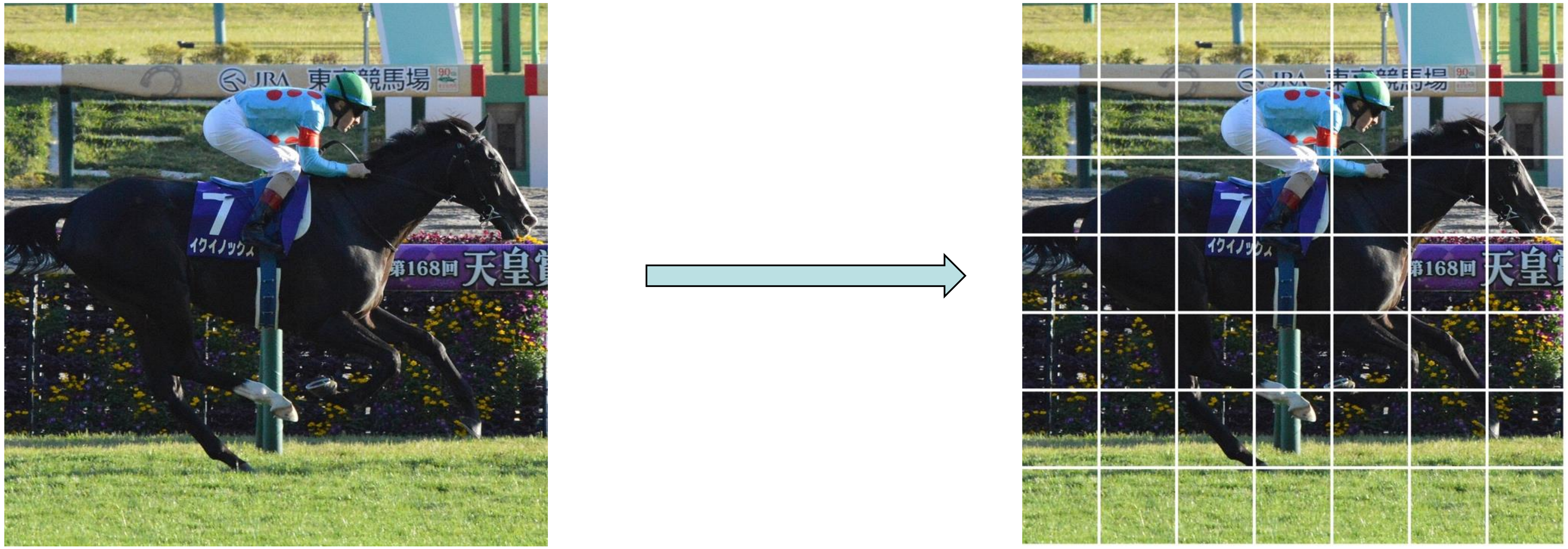
Introduction

Observation

There are noises and redundancies in image tokens:

- Irrelevant backgrounds / less informative tokens
- Similar tokens
- Noises



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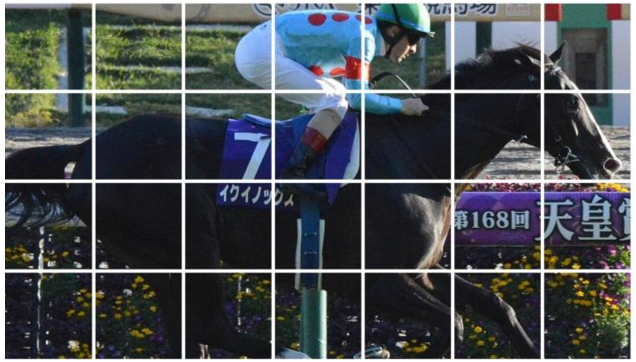

After removing some background tokens, ViT can still classify this image as a horse.

Introduction

Observation

There are noises and redundancies in image tokens:

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Even with a more aggressive removal, a ViT model might still be able to recognise the image.

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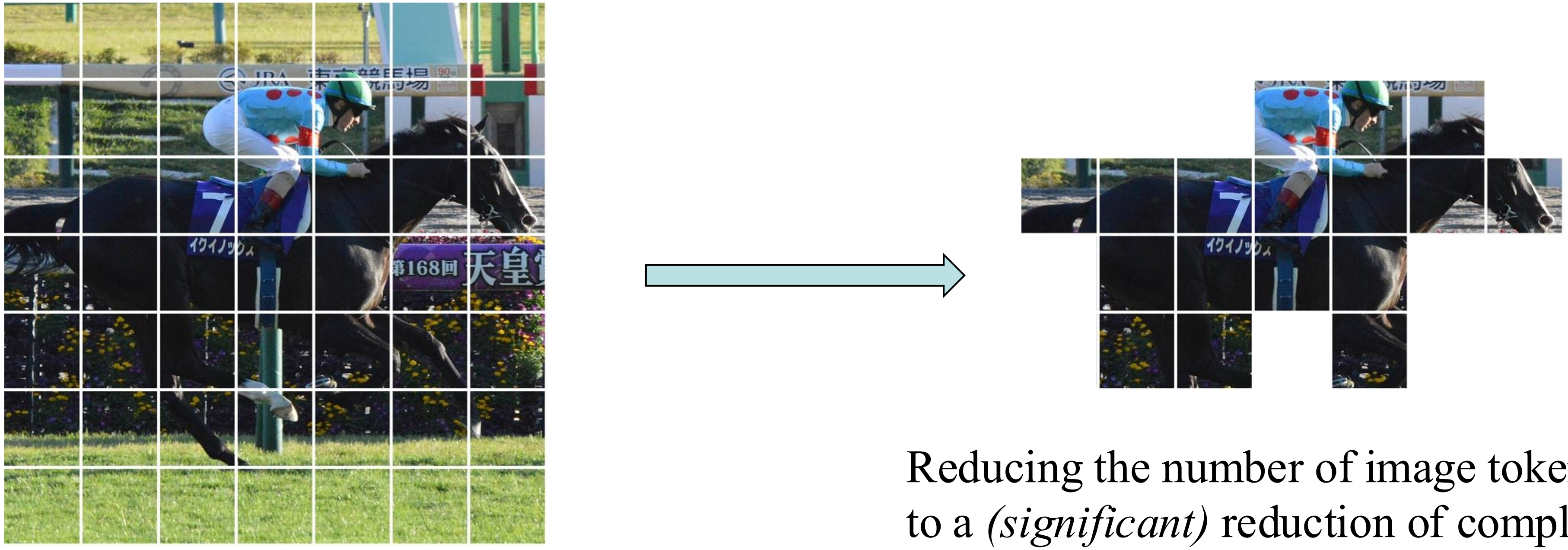


Reducing the number of image tokens (N) leads to a *(significant)* reduction of complexity

Token Reduction

Problem Definition

Given a set of inputs $X = \{x_1, x_2, \dots, x_n\}$, reduce X to another set $Z = \{z_1, z_2, \dots, z_m\}$, where $n \ll m$. The reduced token set Z should preserve input information and model performance as much as possible.

Token Reduction

Methodology Categories

- Token pruning
- Token merging
- Token resampling
- Token routing

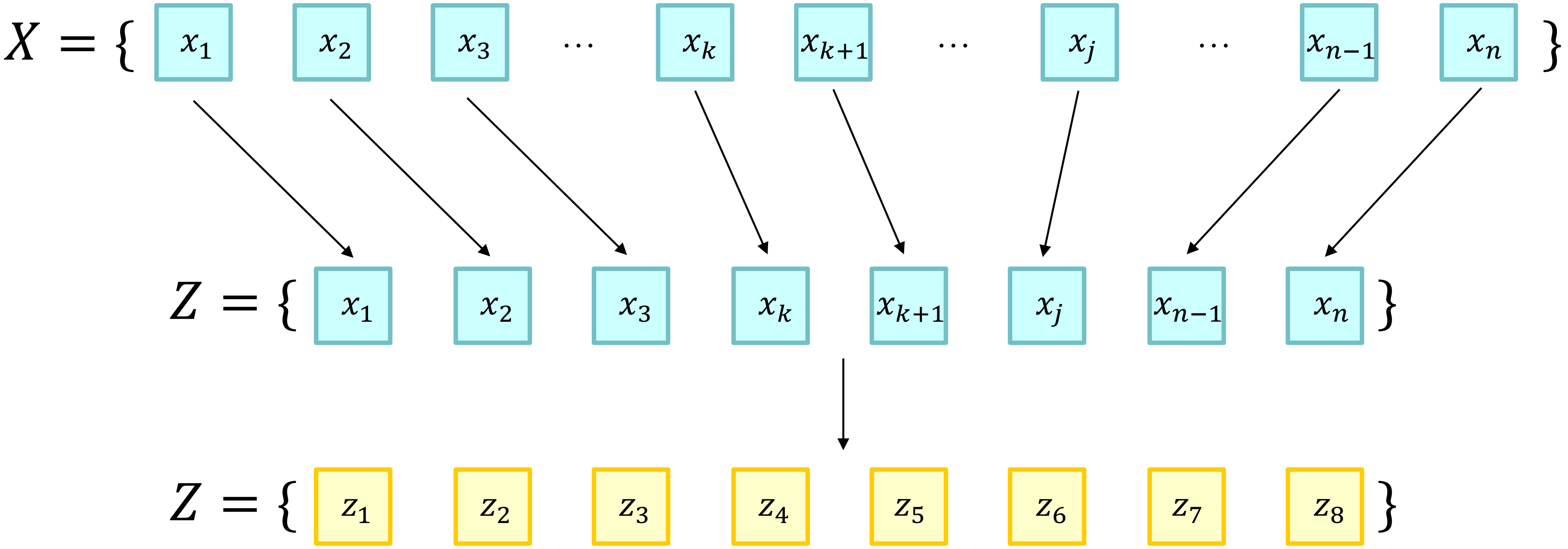
Two types of token reduction:

- Static reduction: Pre-defined reduction pattern for all inputs
- Dynamic/Adaptive reduction: Reduction determined w.r.t. the inputs

Token Reduction

1. Token Pruning

Reduced token set Z is a subset of the original token set X



Token Reduction

1. Token Pruning

Reduced token set Z is a subset of the original token set X

- Assign/Measure importance score for each token and keep only the top-K tokens.
- Importance score can come from attention weights, gradients, learned gates, etc.

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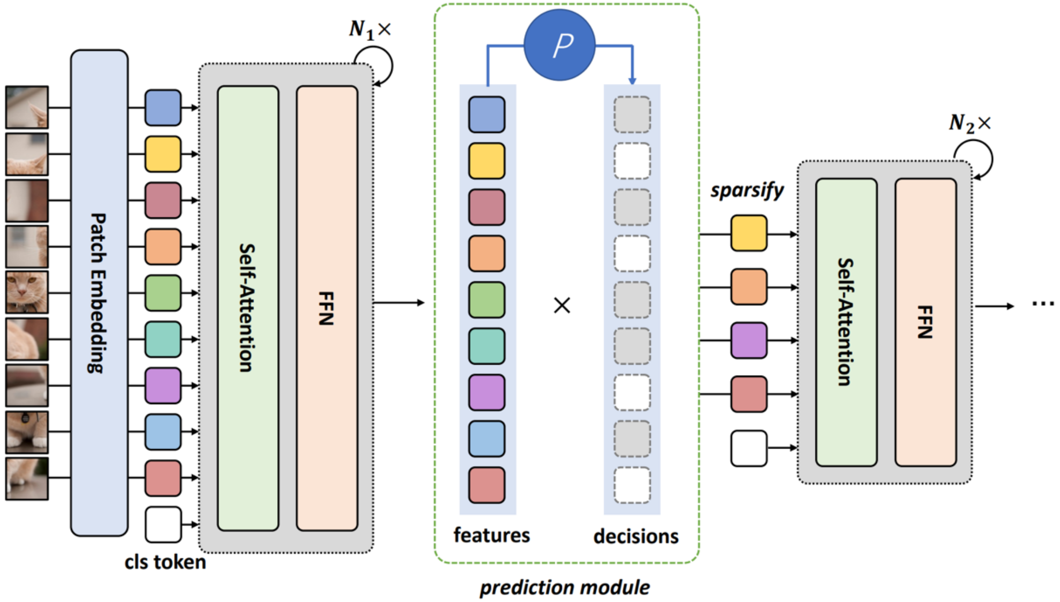

DynamicViT[6]

Apply an MLP-based predictor module to assign token importance score.

$$x_i^{\text{local}} = \text{MLP}(x_i) \in \mathbb{R}^{1\times\frac{C}{2}}$$

$$x^{\text{global}} = \text{MLP}(x_i) \in \mathbb{R}^{1\times\frac{C}{2}}$$

$$x_i' = \left[x_i^{\text{local}}, x^{\text{global}}\right] \in \mathbb{R}^{1\times C}$$

$$\pi_i = \text{softmax}\big(\text{MLP}(x_i')\big) \in \mathbb{R}^{1\times 2}$$

[6] Rao, Yongming, et al. "Dynamicvit: Efficient vision transformers with dynamic token sparsification." In NeurIPS, 2021.

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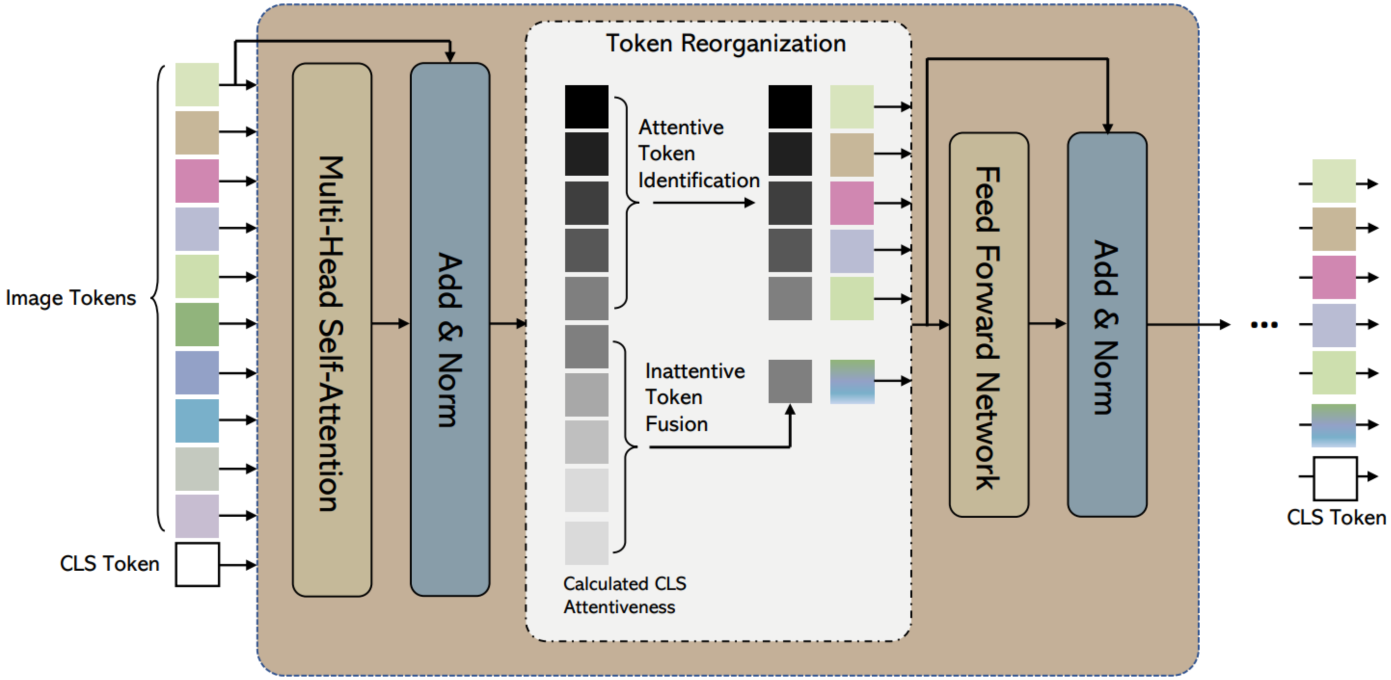

EViT[7]

Use [CLS] attention to determine token importance.

$$\pi_i = \frac{1}{H}\sum_{h=[1,\ldots,H]} A^h_{0,i}$$

[7] Liang, Youwei, et al. "Not all patches are what you need: Expediting vision transformers via token reorganizations." In ICLR, 2022.

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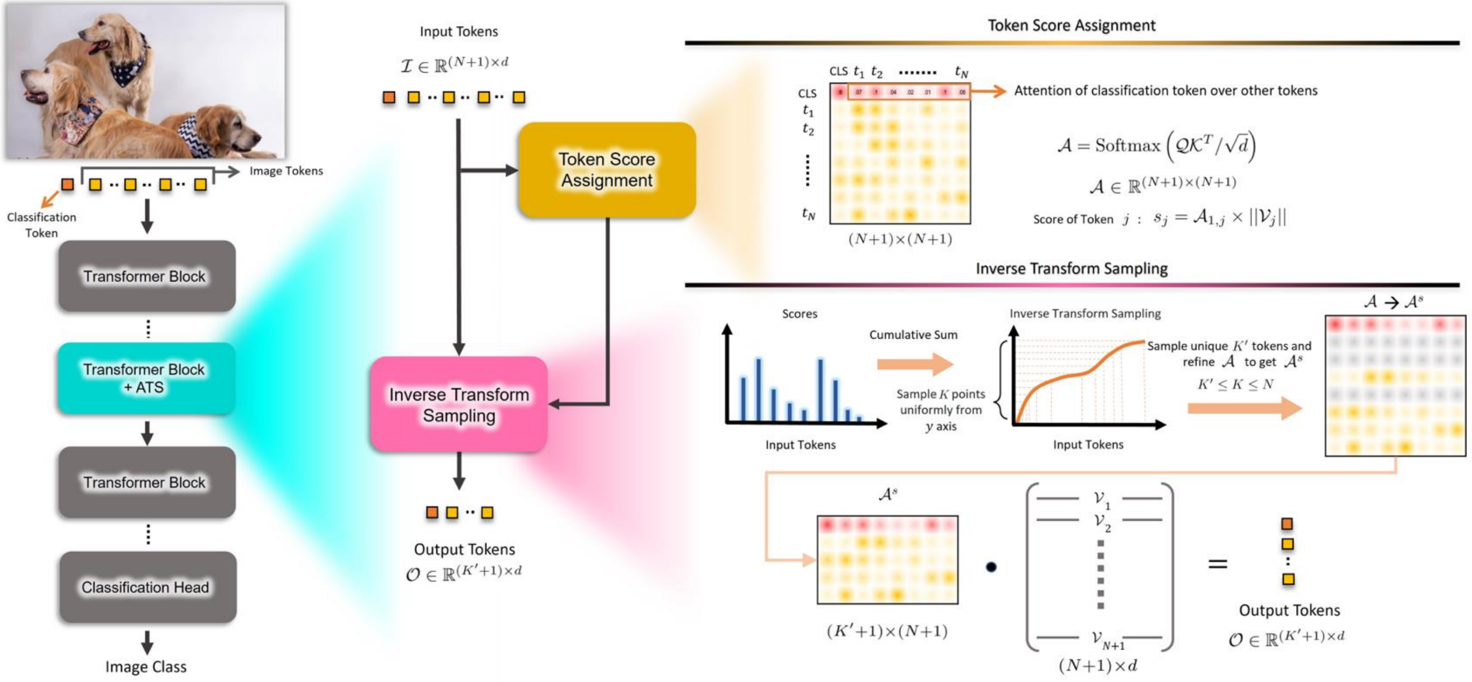

ATS[8]

Maintain token score and leverage Inverse Transform Sampling on the score.

$$S_i = \frac{A_{0,i} \times \|V_i\|}{\sum_{j=1} A_{0,j} \times \|V_j\|}$$

$$\mathrm{CDF}_i = \sum_{j=1}^{j=i} S_j$$

$$\psi(k) = \mathrm{CDF}^{-1}(k)$$

[8] Fayyaz, Mohsen, et al. "Adaptive token sampling for efficient vision transformers." In ECCV, 2022.

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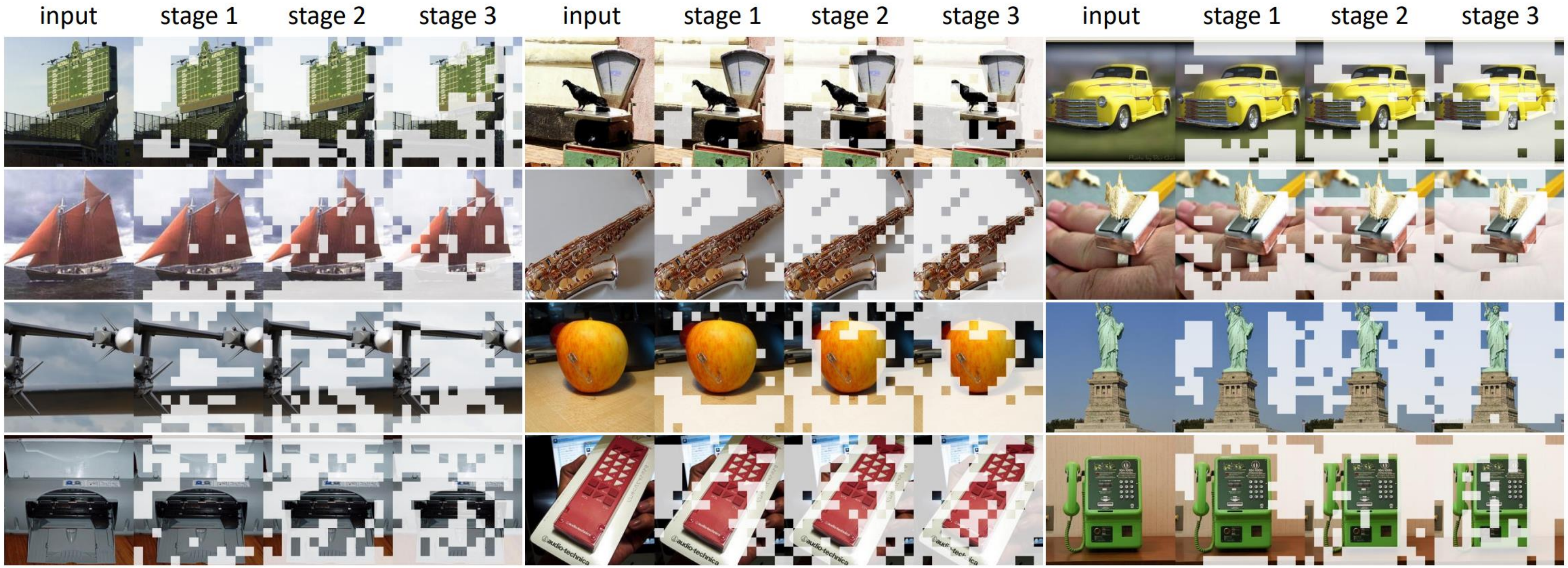


Token pruning outcome examples

[6] Rao, Yongming, et al. "Dynamicvit: Efficient vision transformers with dynamic token sparsification." In NeurIPS, 2021.

Token Reduction

1. Token Pruning

Reduced token set Z is a subset of the original token set X

- Assign/Measure importance score for each token and keep only the top-K tokens.
- Importance score can come from attention weights, gradients, learned gates, etc.

Main advantages are

- Straightforward and consistent with human intuition
- Easy to interpret and visualise

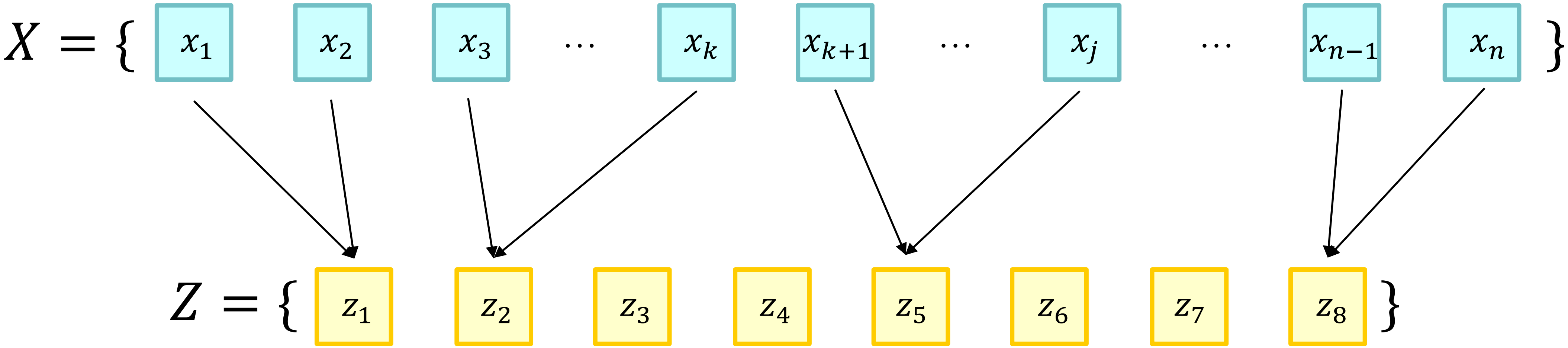
Main drawbacks are

- Irreversible information loss from hard decisions
- Finetuning or retraining is often required to recover performance

Token Reduction

2. Token Merging

Reduced token set Z consists of merged tokens from the original token set X



Token Reduction

2. Token Merging

Reduced token set Z consists of merged tokens from the original token set X

- Instead of dropping tokens, merge similar ones into a fused representative.
- Merging can be based on clustering, cosine similarity, or local adjacency.

Token Reduction

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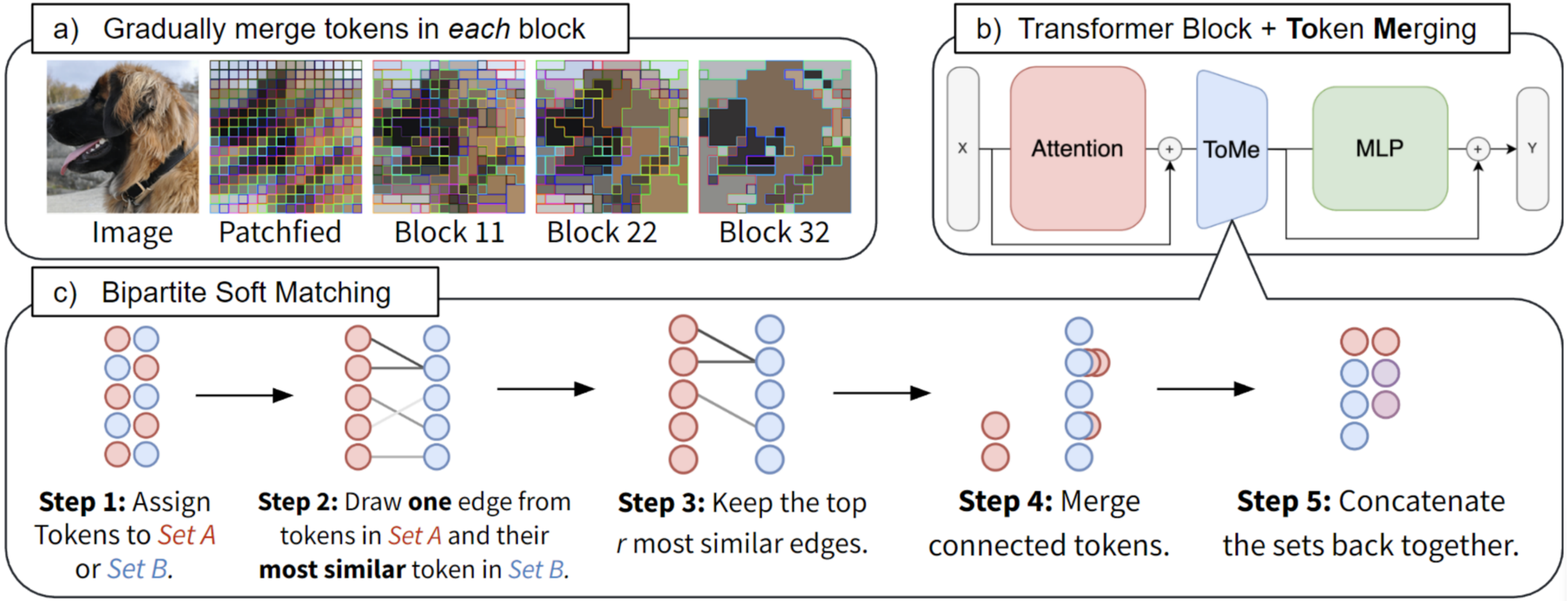

ToMe[9]

[9] Bolya, Daniel, et al. "Token merging: Your vit but faster." In ICLR, 2023.

Token Reduction

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- Instead of dropping tokens, merge similar ones into a fused representative.
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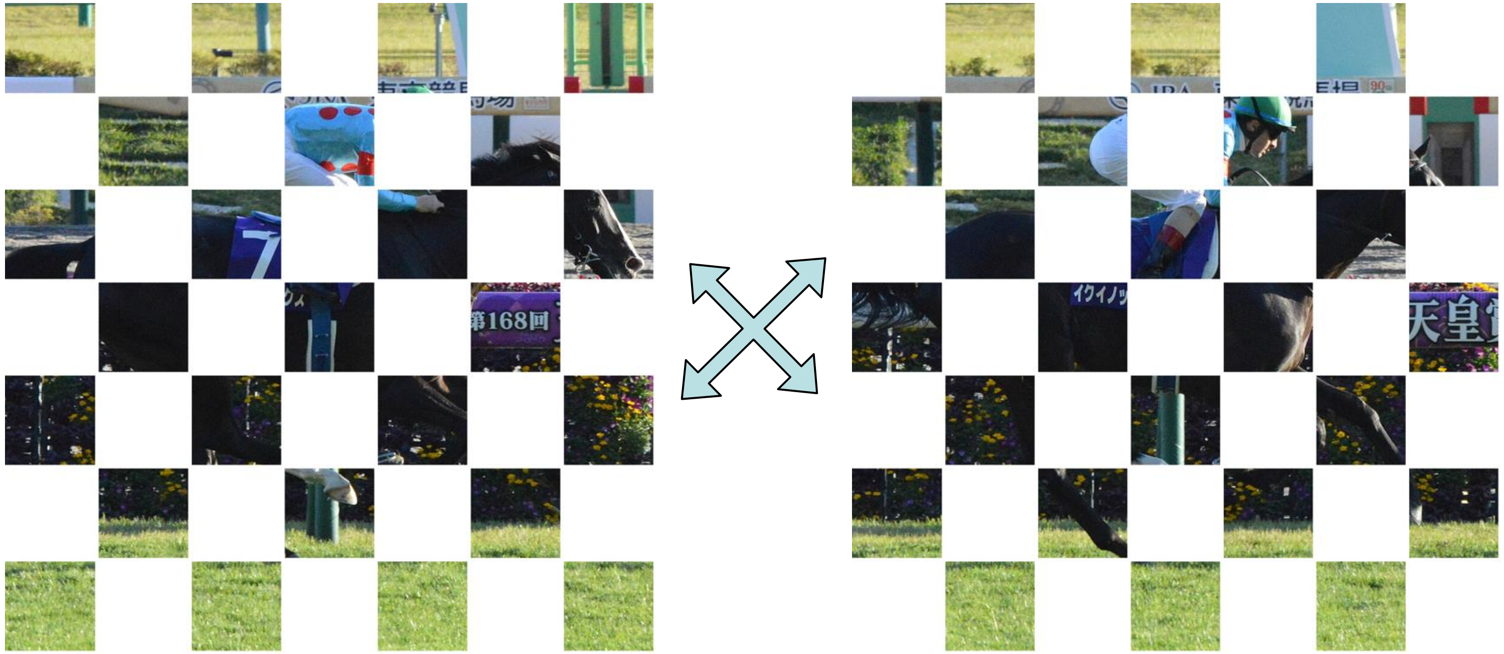
Step 1: The tokens are first split into two groups. Cosine similarity is then computed for each pair of tokens drawn from different groups.



Group A

Group B

Token Reduction

2. Token Merging

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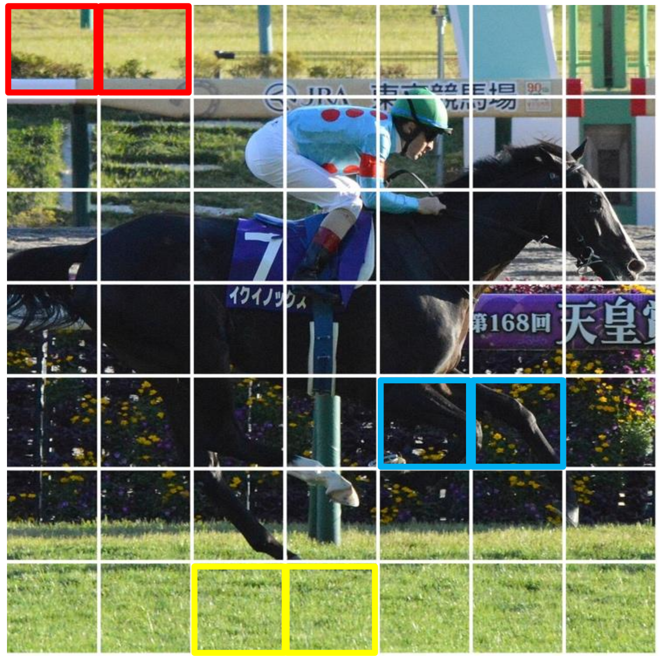
Step 2: Identify the top-K similar pairs of tokens.



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Step 3: Merge the selected pairs of similar tokens with weights, and then replace the original tokens with the merged tokens

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Step 4: Update the token weights to reflect the importance/volume of their information

$$A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \log(s)\right)$$

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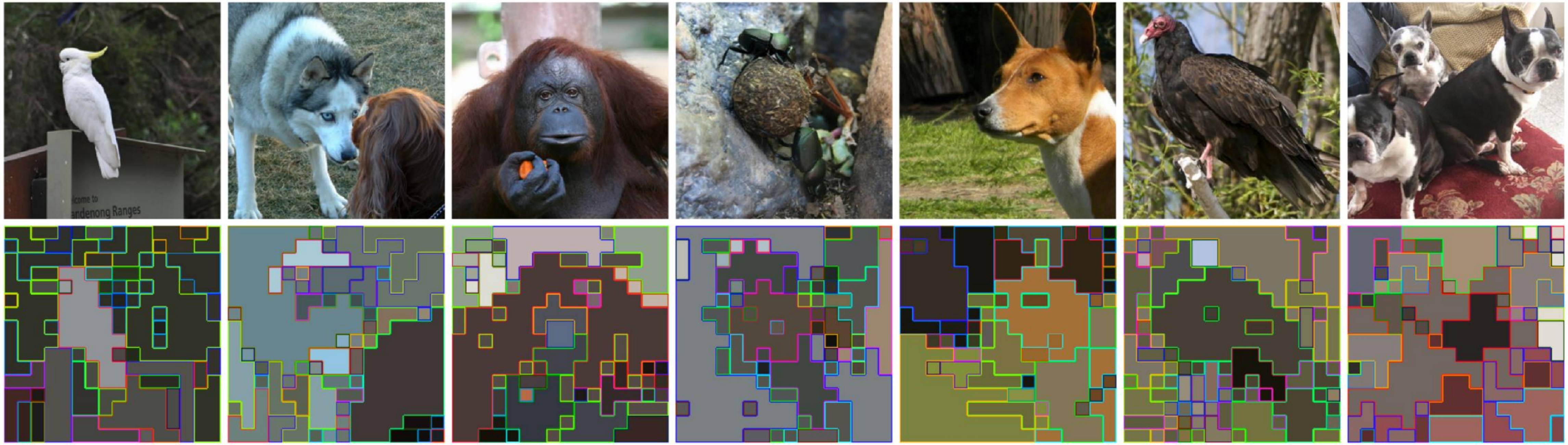

Token merging outcome examples

[9] Bolya, Daniel, et al. "Token merging: Your vit but faster." In ICLR, 2023.

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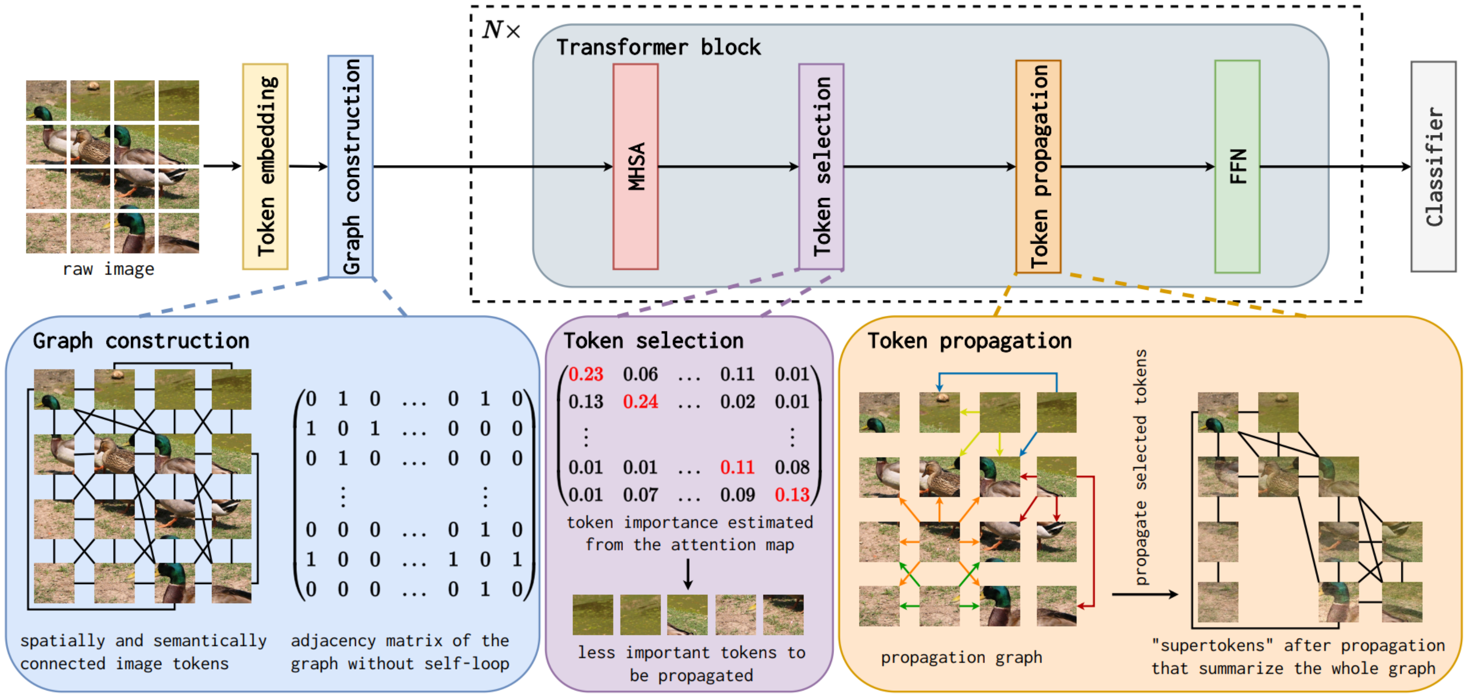

GTP-ViT[10]

Merge tokens according to both spatial and semantic relationships.
Graph-based token fusion

$$Z = X^k + \alpha A^p X^p$$

[10] Xu, Xuwei, et al. "Gtp-vit: Efficient vision transformers via graph-based token propagation." In WACV, 2024.

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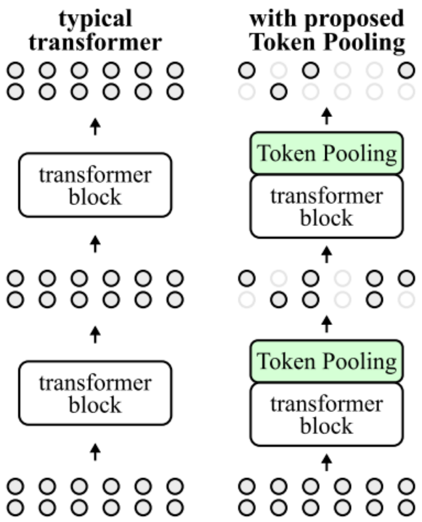
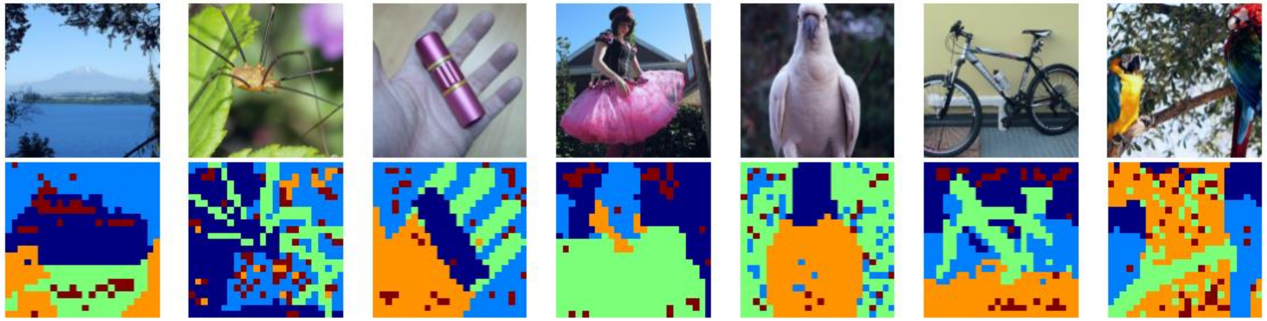


TPS[11]



Merge tokens based on K-Medoids clustering

Token Reduction

2. Token Merging

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- Merging can be based on clustering, cosine similarity, or local adjacency.

Main advantages are

- Yield smoother performance degradation than pruning
- Preserve necessary information
- Usually finetuning-free

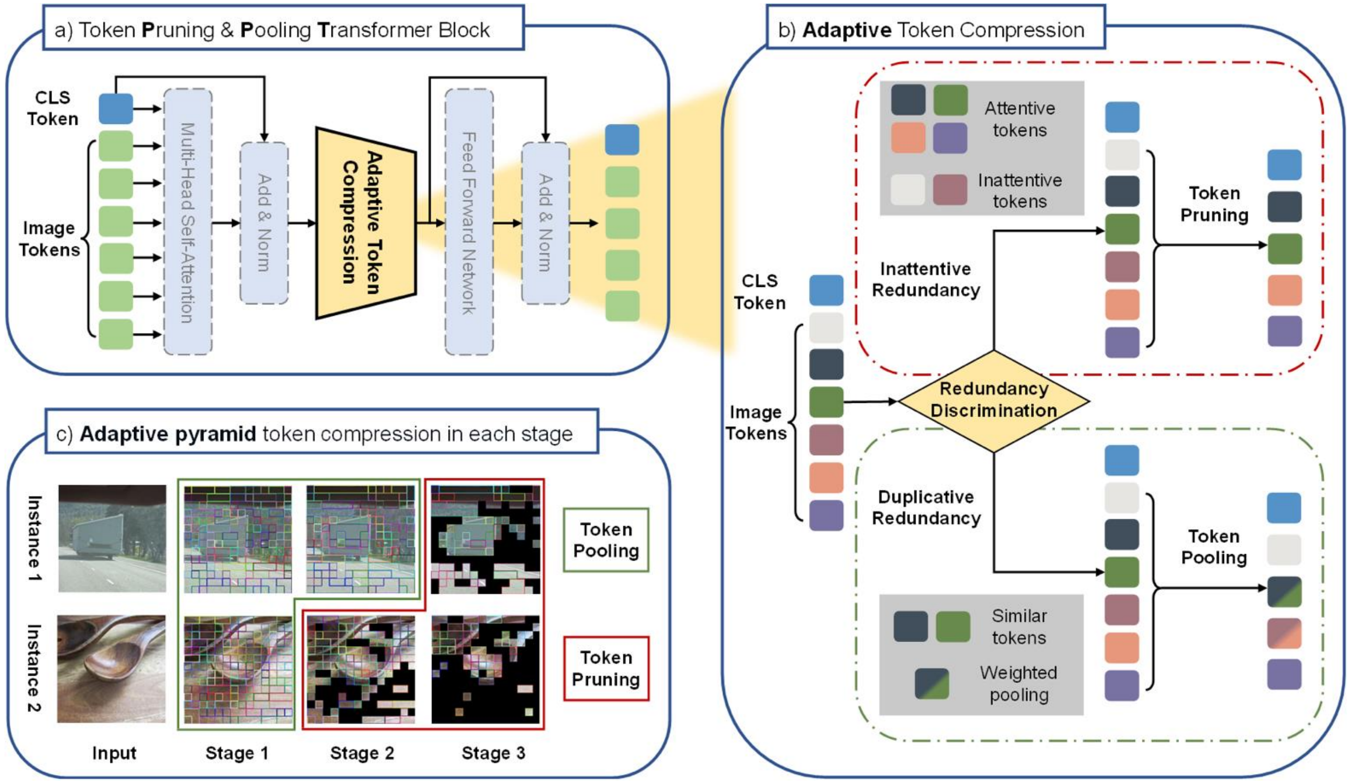
Main drawbacks are

- Merging may blur distinctions that matter for fine-grained tasks
- Identifying similar tokens can be time-consuming for long sequence lengths

Token Reduction

1+2. Token Pruning + Token Merging

Dynamically determine if the token should be pruned or merged



PPT[12]

Token pruning for inattentive tokens, which are most likely noises or irrelevant backgrounds. Token merging for duplicative tokens to prevent redundant information.

[12] Wu, Xinjian, et al. "Ppt: Token pruning and pooling for efficient vision transformers." In WACV, 2024.

Token Reduction

1+2. Token Pruning + Token Merging

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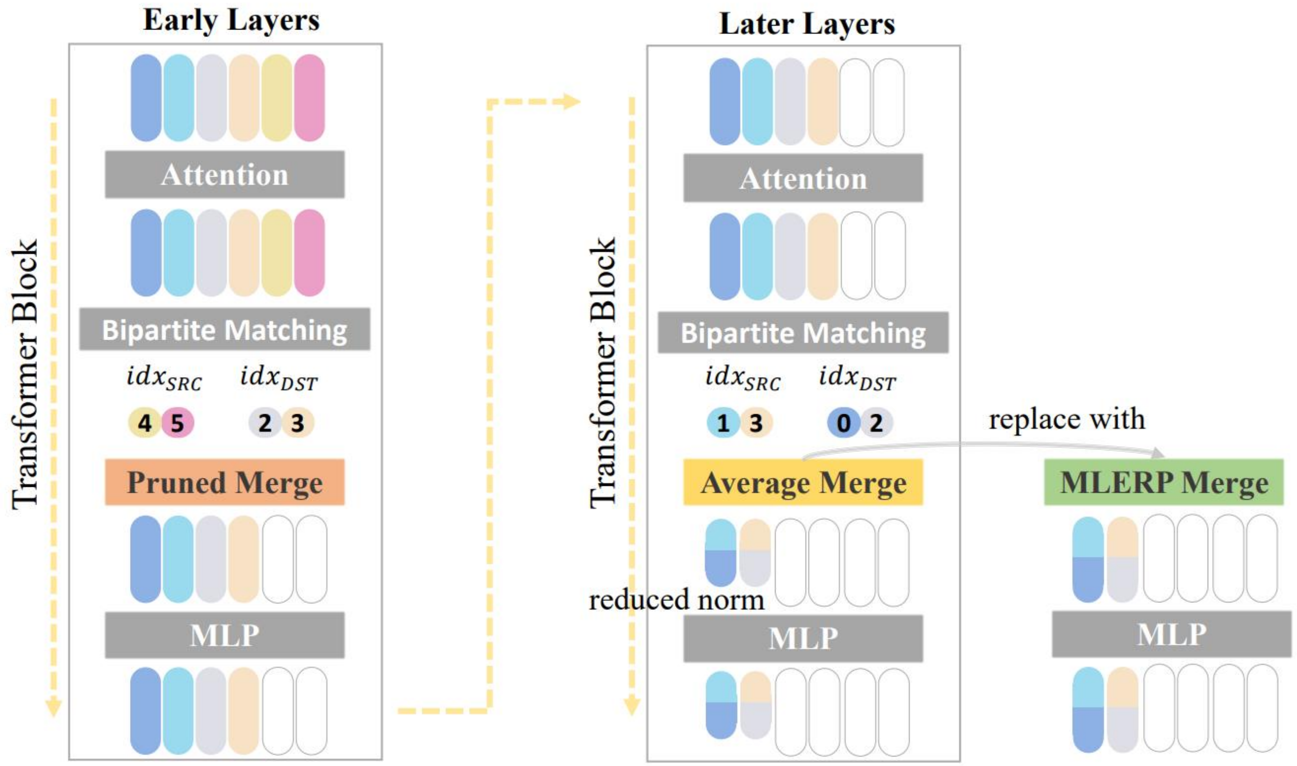


ToFu[13]

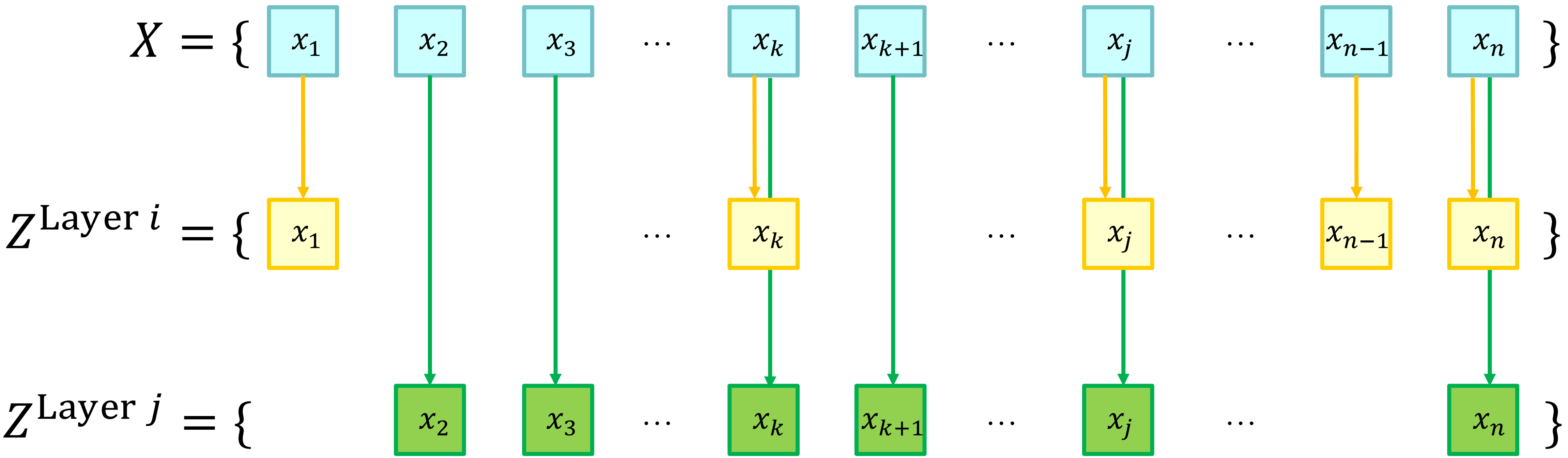
Token pruning for shallow layers that extracts coarse features. Token merging for deep layers that focus on fine-grained hidden patterns.

[13] Kim, Minchul, et al. "Token fusion: Bridging the gap between token pruning and token merging." In WACV, 2024.

Token Reduction

3. Token Routing

Reduced token set Z can include previously unselected tokens



Token Reduction

3. Token Routing

Reduced token set Z can include previously unselected tokens

- A learned router or a heuristic algorithm evaluates tokens on the fly and decides their computational path.
- Selection can depend on content, confidence, uncertainty estimates, etc.

(Broadly, token pruning and token merging can be viewed as two distinct routing methods. They are special cases in token routing where the tokens would never be re-used. Here, we only discuss about cases when dynamic computation paths allow tokens to be re-used.)

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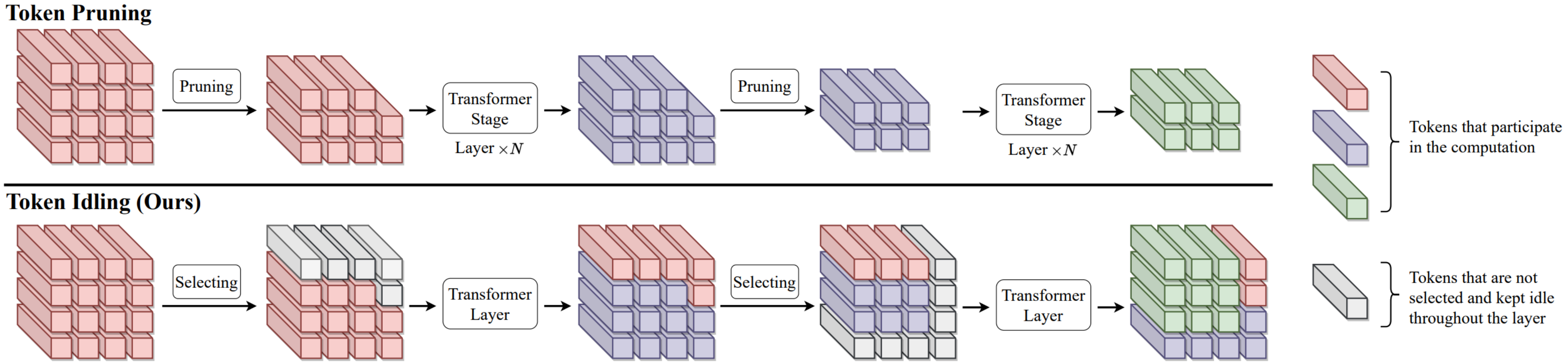

IdleViT[14]

Re-select tokens that are regarded inattentive in shallow layers.

[14] Xu, Xuwei, et al. "No token left behind: Efficient vision transformer via dynamic token idling." In AJCAI, 2023.

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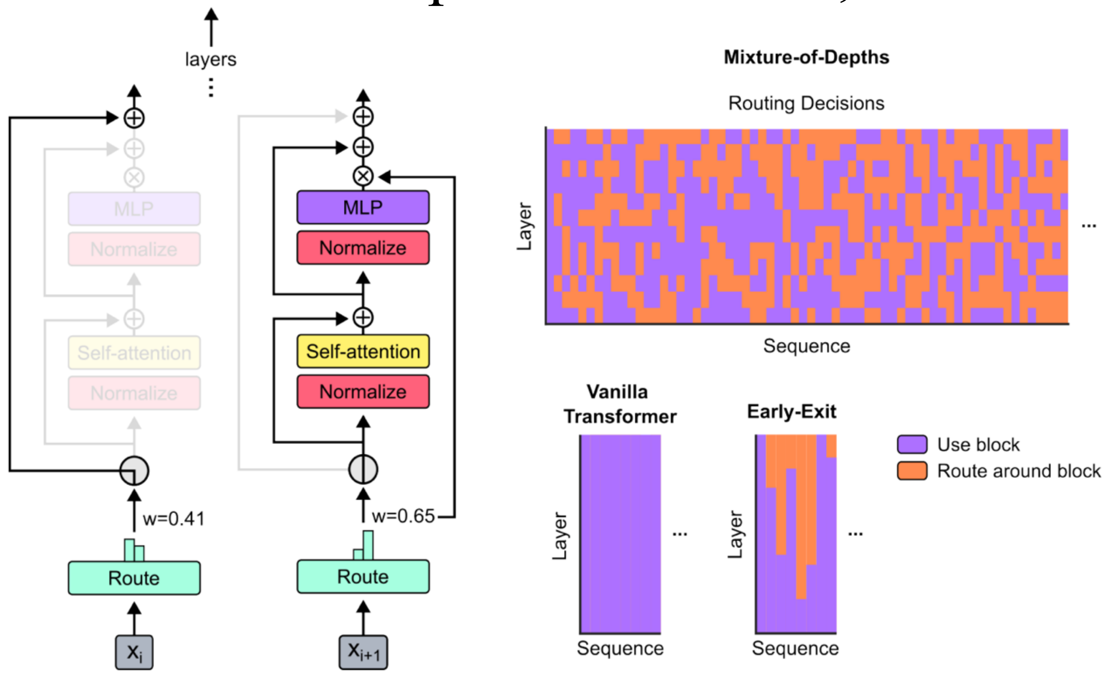

MoD[15]

An efficient router at every layer determines whether each token should bypass the current layer or participate in the computation.

[15] Raposo, David, et al. "Mixture-of-depths: Dynamically allocating compute in transformer-based language models." arXiv:2404.02258, 2024.

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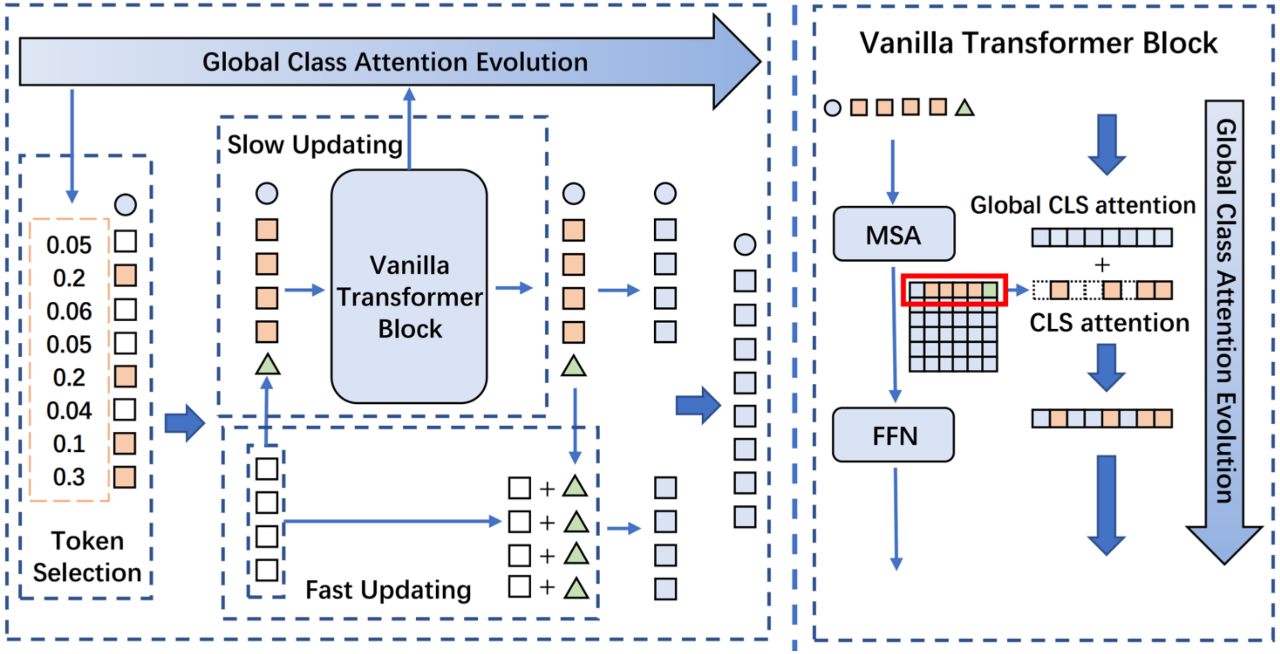

Evo-ViT[16]

Tokens are selected into two groups using accumulated attention weights, one of which participates in the self-attention computation while the other group leverages fast updating.

$$A_{\mathrm{cls},g}^{l} = \alpha A_{\mathrm{cls},g}^{l-1} + (1-\alpha)\, A_{\mathrm{cls}}^{l}$$

[16] Xu, Yifan, et al. "Evo-vit: Slow-fast token evolution for dynamic vision transformer." In AAAI, 2022.

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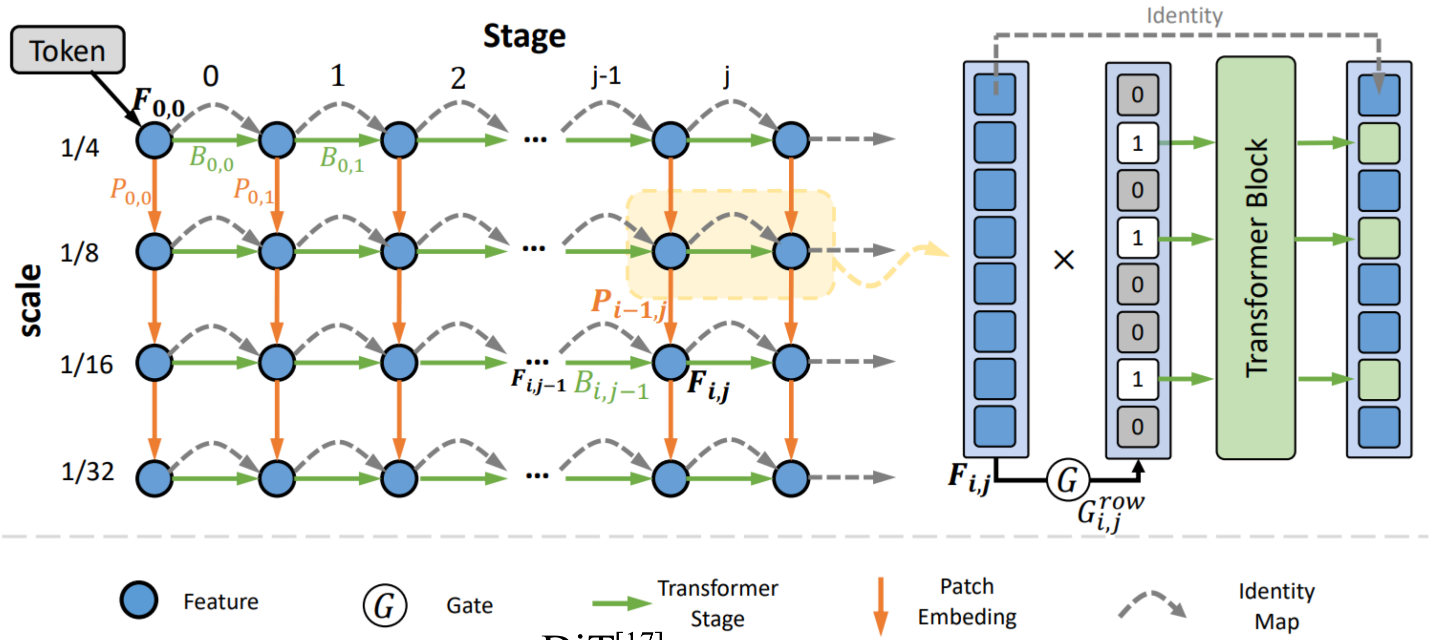

DiT[17]

Tokens are dynamically routed through a two-dimensional computation graph. A token is selected to normally undergo the self-attention, or bypass the current layer, or aggregate with the other tokens.

[17] Ma, Yuchen, Zhengcong Fei, and Junshi Huang. "DiT: Efficient vision transformers with dynamic token routing." arXiv:2308.03409, 2023.

Token Reduction

3. Token Routing

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Main advantages are

- Neither completely remove information nor completely blur information
- Allow more dynamic computing resources allocation

Main drawbacks are

- Increased hallucination possibility
- Additional memory overhead for cached tokens

Token Reduction

4. Token Resampling

Reduced token set Z learns from the original token set X

$$X = \{\ \boxed{x_1}\ \ \boxed{x_2}\ \ \boxed{x_3}\ \cdots\ \boxed{x_k}\ \ \boxed{x_{k+1}}\ \cdots\ \boxed{x_j}\ \cdots\ \boxed{x_{n-1}}\ \ \boxed{x_n}\ \}$$

$$+$$

$$Q = \{\ \boxed{q_1}\ \ \boxed{q_2}\ \ \boxed{q_3}\ \ \boxed{q_4}\ \ \boxed{q_5}\ \ \boxed{q_6}\ \ \boxed{q_7}\ \ \boxed{q_8}\ \}$$

$$\downarrow$$

$$Z = \{\ \boxed{z_1}\ \ \boxed{z_2}\ \ \boxed{z_3}\ \ \boxed{z_4}\ \ \boxed{z_5}\ \ \boxed{z_6}\ \ \boxed{z_7}\ \ \boxed{z_8}\ \}$$

Token Reduction

4. Token Resampling

Reduced token set Z learns from the original token set X

- Learn an explicit map from many input tokens to a smaller latent token set.
- Representative examples include latent arrays and autoencoder-style bottlenecks.

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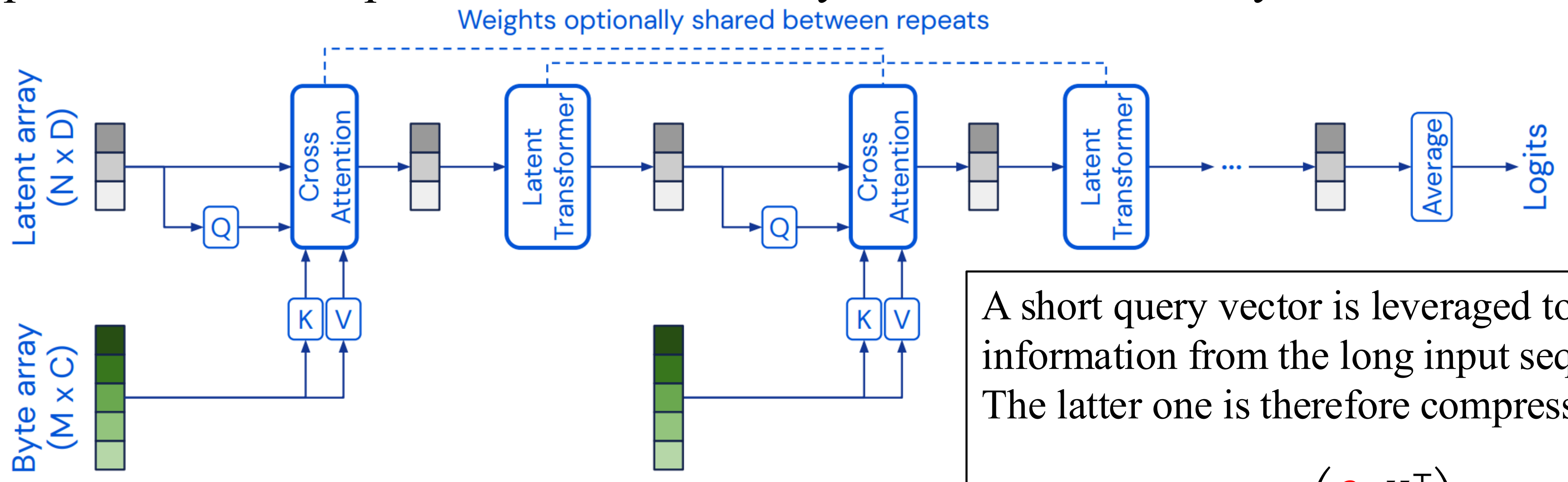

Perceiver[18]

A short query vector is leveraged to extract information from the long input sequence. The latter one is therefore compressed.

$$Z = \mathrm{softmax}\left(\frac{Q_q K_X^\top}{\sqrt{d_k}}\right) V_X + Q_q$$

[18] Jaegle, Andrew, et al. "Perceiver: General perception with iterative attention." In ICML, 2021.

Token Reduction

4. Token Resampling

Reduced token set *Z* learns from the original token set *X*

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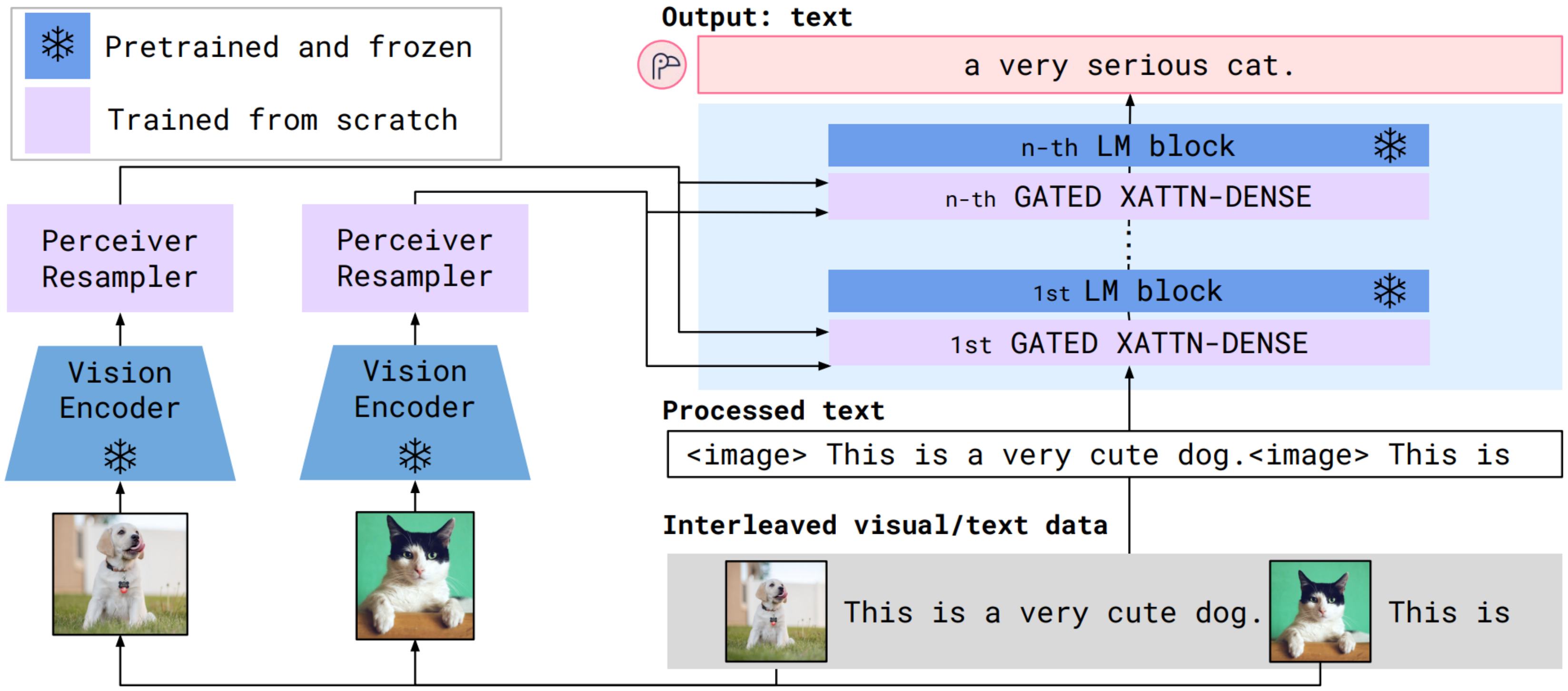

Flamingo[19]

A Perceiver token resampler for vision-language model that reduces the number of input image tokens to only 64.

[19] Alayrac, Jean-Baptiste, et al. "Flamingo: a visual language model for few-shot learning." In NeurIPS, 2022.

Token Reduction

4. Token Resampling

Reduced token set Z learns from the original token set X

- Learn an explicit map from many input tokens to a smaller latent token set.
- Representative examples include latent arrays and autoencoder-style bottlenecks.

Main advantages are

- High compression ratio
- End-to-end trainable and often preserves semantics well

Main drawbacks are

- Trade-off of extra architectural complexity and parameters
- Need to be finetuned or retrained

Token Reduction

Information Bottleneck[20]

Information bottleneck (IB) serves as a theoretical foundation for token reduction:

- IB method formalizes the trade-off between compression and prediction.
- The objective is to extract a compressed representation (Z) from an input (X) that retains the maximum possible information about a target output (Y):

$$\min_{Z}\big(I(X;Z) - \beta I(Z;Y)\big)$$

$I(X;Z)$: Mutual information between input and compressed state (minimize this to compress).

$I(Z;Y)$: Mutual information between compressed state and target (maximize this to retain accuracy).

β: The Lagrange multiplier controlling the trade-off.

[20] Tishby, Naftali, Fernando C. Pereira, and William Bialek. "The information bottleneck method." arXiv preprint physics/0004057, 2000.

Token Reduction

Information Bottleneck[20]

Different token reduction methods from the bottleneck perspective:

Method	IB interpretation
Pruning	Hard bottleneck with discrete retention decisions. It treats the reduction process as a binary selection problem where $\subset X$
Merging	Soft bottleneck with weighted aggregation or clustering. Instead of discarding information, it performs a local lossy compression by mapping multiple input points X_i, X_j to a single representation Z_k.
Routing	Adaptive bottleneck conditioned on the input. Not all tokens in X require the same amount of information processing to predict Y. It treats computation as a finite resource. A router learns to predict which tokens are "information-rich" and require the full depth of the transformer, and which are "predictable" and can bypass layers.
Resampling	Latent bottleneck with a shorter latent query vector. Token resampling is often used in long-context or multi-modal models to select a fixed-size representation from a variable-length input. It attempts to find a distribution of Z that best approximates the density of the most relevant features in X.

[20] Tishby, Naftali, Fernando C. Pereira, and William Bialek. "The information bottleneck method." arXiv preprint physics/0004057, 2000.

Token Reduction

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Information Bottleneck[20]

Different token reduction methods from the information strategy perspective:

Method	Information strategy
Pruning	Elimination
Merging	Aggregation
Routing	Adaptation
Resampling	Summarisation

[20] Tishby, Naftali, Fernando C. Pereira, and William Bialek. "The information bottleneck method." arXiv preprint physics/0004057, 2000.

Thank you!

Q&A